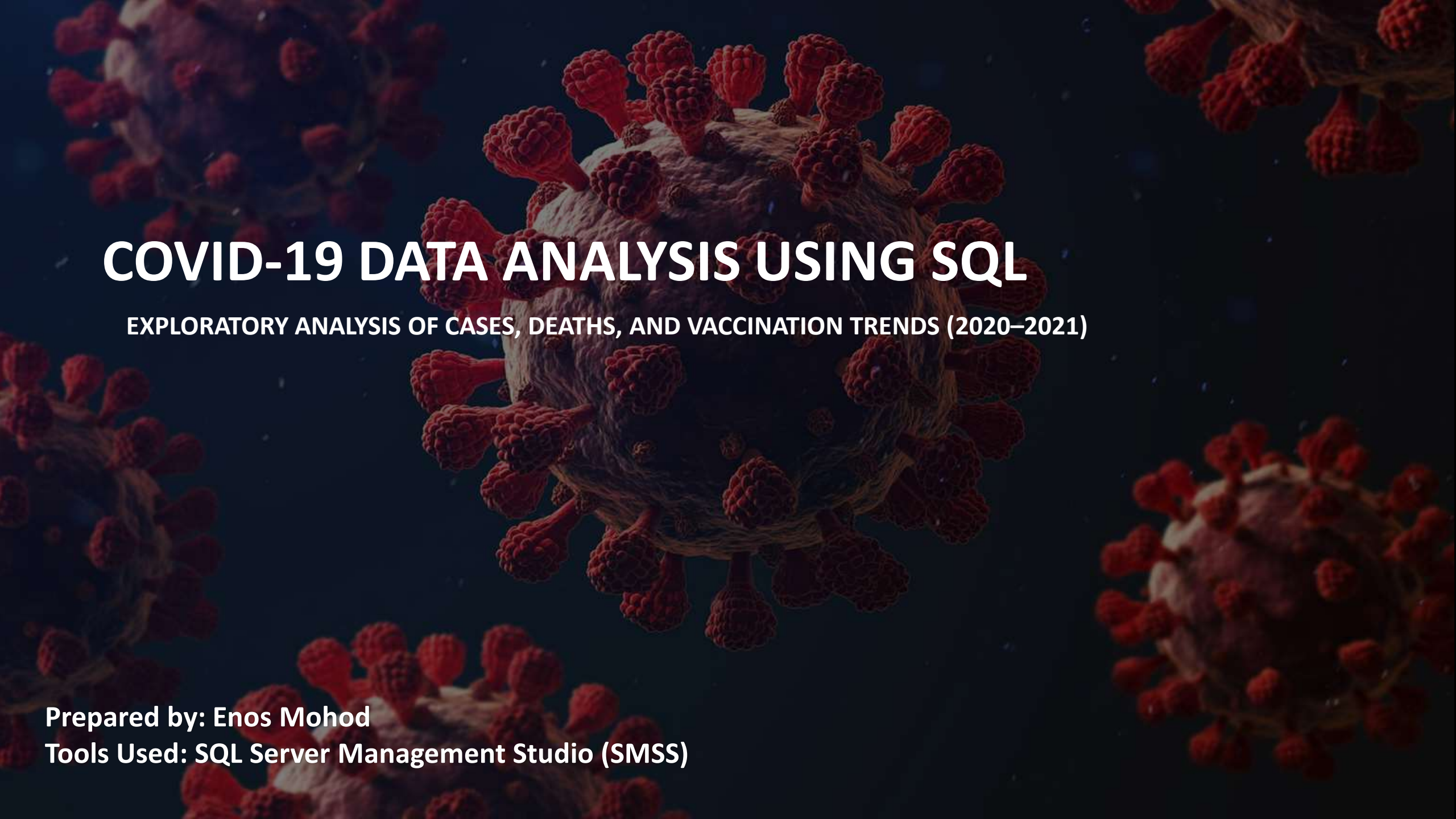




COVID-19 DATA ANALYSIS USING SQL

EXPLORATORY ANALYSIS OF CASES, DEATHS, AND VACCINATION TRENDS (2020–2021)

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Tools Used: SQL Server Management Studio (SMSS)

PROBLEM STATEMENT & OBJECTIVES

PROBLEM STATEMENT

THE COVID-19 PANDEMIC GENERATED LARGE VOLUMES OF GLOBAL DATA RELATED TO CASES, DEATHS, AND VACCINATIONS. HOWEVER, RAW DATA ALONE DOES NOT CLEARLY EXPLAIN:

- HOW THE IMPACT VARIED ACROSS REGIONS
- HOW SEVERITY CHANGED OVER TIME
- HOW VACCINATION PROGRESSED RELATIVE TO POPULATION SIZE

THIS ANALYSIS AIMS TO CONVERT RAW COVID DATA INTO MEANINGFUL, STRUCTURED INSIGHTS USING SQL.

Objectives

- Analyze COVID **cases and deaths** across countries and continents
- Identify **time-based trends** in death severity
- Compare **population-normalized metrics** to avoid misleading absolute numbers
- Examine **vaccination rollout trends** using cumulative analysis
- Focus on **descriptive and comparative insights**, not predictions

DATASET OVERVIEW

DATASET DESCRIPTION

- GLOBAL COVID-19 PUBLIC DATASET COVERING **2020–2021**
- CONTAINS DAILY RECORDS FOR MULTIPLE COUNTRIES AND REGIONS

INCLUDES INFORMATION ON:

- CONFIRMED CASES
- DEATHS
- POPULATION
- VACCINATION DATA

KEY CHARACTERISTICS

- **GRANULARITY:** DAILY, COUNTRY-LEVEL DATA
- **GEOGRAPHIC COVERAGE:** GLOBAL (COUNTRIES & CONTINENTS)
- **DATA NATURE:** TIME-SERIES WITH CUMULATIVE AND DAILY METRICS

INSIGHT - 1

COVID DEATH PERCENTAGE VARIES SIGNIFICANTLY ACROSS COUNTRIES

KEY OBSERVATION

WHEN COMPARING **TOTAL DEATHS RELATIVE TO TOTAL CASES** AT THE COUNTRY LEVEL, THERE IS A **WIDE VARIATION** IN DEATH PERCENTAGES ACROSS COUNTRIES.

KEY FINDINGS

- **YEMEN** RECORDS THE **HIGHEST DEATH PERCENTAGE** AT APPROXIMATELY **25.94%**, INDICATING A VERY HIGH FATALITY RATE RELATIVE TO REPORTED CASES.
- **SINGAPORE** RECORDS ONE OF THE **LOWEST DEATH PERCENTAGES** AT APPROXIMATELY **0.05%**, DESPITE HAVING REPORTED CASES.
- **INDIA**, USED AS A COUNTRY-SPECIFIC REFERENCE, SHOWS A DEATH PERCENTAGE OF APPROXIMATELY **1.45%**, WHICH LIES BETWEEN THE GLOBAL EXTREMES.

INSIGHT - 2

COVID DEATH PERCENTAGE IS MORE CONSISTENT AT THE CONTINENT LEVEL

KEY OBSERVATION

WHEN **COVID DEATH** PERCENTAGE IS ANALYZED AT THE **CONTINENT LEVEL**, THE VARIATION IS **MUCH SMALLER** COMPARED TO COUNTRY-LEVEL ANALYSIS.

KEY FINDINGS

- THE **EUROPEAN UNION** SHOWS THE **HIGHEST DEATH PERCENTAGE** AT APPROXIMATELY **2.83%**.
- **AFRICA, NORTH AMERICA, SOUTH AMERICA, EUROPE, AND OCEANIA** ALL SHOW **SIMILAR DEATH PERCENTAGES**, GENERALLY ABOVE **~2.5%**.
- **ASIA** RECORDS THE **LOWEST DEATH PERCENTAGE** AT APPROXIMATELY **1.66%**.

INSIGHT - 3

GLOBAL COVID DEATH PERCENTAGE TREND OVER TIME

KEY OBSERVATION

WHEN ANALYZING **COVID DEATH PERCENTAGE GLOBALLY** OVER TIME, A CLEAR PEAK IN SEVERITY IS OBSERVED DURING THE **MIDDLE OF 2020**, FOLLOWED BY A **GRADUAL DECLINE**.

KEY FINDINGS

- DURING THE **MIDDLE MONTHS OF 2020**, GLOBAL DEATH PERCENTAGE WAS **SIGNIFICANTLY HIGH**, REACHING APPROXIMATELY **6%–7%**, INDICATING A SEVERE PHASE OF THE PANDEMIC.
- AFTER MID-2020, THE DEATH PERCENTAGE BEGAN TO **DECLINE STEADILY**, DESPITE INCREASING CASE VOLUMES.
- BY **LATE 2020 AND EARLY 2021**, GLOBAL DEATH PERCENTAGE STABILIZED AROUND **~2%–2.5%**.

INSIGHT - 4

INDIA: COVID DEATH PERCENTAGE TREND OVER TIME

KEY OBSERVATION

INDIA EXPERIENCED ITS **HIGHEST COVID DEATH PERCENTAGE DURING MID-2020**, AFTER WHICH THE RATE SHOWED A **STEADY AND SUSTAINED DECLINE INTO 2021**.

KEY FINDINGS

- IN EARLY 2020, INDIA'S DEATH PERCENTAGE WAS **RELATIVELY LOW (~1.6%)**.
- **DURING APRIL–JUNE 2020**, THE DEATH PERCENTAGE INCREASED SHARPLY, **PEAKING AROUND ~3.6%**, ALIGNING WITH THE **GLOBAL SEVERITY PEAK**.
- **AFTER MID-2020**, THE DEATH PERCENTAGE CONSISTENTLY DECLINED, FALLING TO **APPROXIMATELY ~1.1% BY APRIL 2021**.
- ONCE THE **DOWNWARD TREND BEGAN**, NO FURTHER SHARP INCREASES WERE OBSERVED.

INSIGHT - 5

POPULATION-NORMALIZED COVID INFECTION COMPARISON (COUNTRY LEVEL)

KEY OBSERVATION

WHEN **TOTAL COVID CASES** ARE COMPARED AGAINST POPULATION SIZE, A SMALL GROUP OF COUNTRIES SHOWS DISPROPORTIONATELY **HIGH INFECTION PENETRATION**.

KEY FINDINGS

- **ANDORRA** RECORDS THE **HIGHEST** POPULATION INFECTION RATE, WITH **APPROXIMATELY ~17.12%** OF ITS POPULATION REPORTED AS INFECTED.
- OTHER COUNTRIES SUCH AS **MONTENEGRO, CZECHIA, SAN MARINO, AND LUXEMBOURG** ALSO SHOW HIGH INFECTION PERCENTAGES, GENERALLY IN THE **~12%–15%** RANGE.
- **MOST COUNTRIES** FALL WELL BELOW THESE LEVELS, HIGHLIGHTING HOW EXTREME THESE CASES ARE IN PROPORTIONAL TERMS.

INSIGHT - 6

INDIA: POPULATION-NORMALIZED COVID INFECTION TREND

KEY OBSERVATION

WHEN TOTAL COVID CASES ARE ANALYZED AS A PERCENTAGE OF INDIA'S POPULATION, THE INFECTION RATE INCREASED OVER TIME BUT REMAINED BELOW 2%, EVEN AT ITS PEAK.

KEY FINDINGS

- IN EARLY 2020, THE PERCENTAGE OF POPULATION INFECTED IN INDIA WAS **NEGLIGIBLE (NEAR 0%)**.
- THE INFECTION PERCENTAGE **GRADUALLY INCREASED OVER TIME**, REFLECTING THE SPREAD OF CASES ACROSS THE COUNTRY.
- THE **HIGHEST POPULATION INFECTION PERCENTAGE** WAS OBSERVED IN **APRIL 2021**, AT APPROXIMATELY **~1.38%**.
- EVEN AT ITS PEAK, INDIA'S POPULATION INFECTION RATE REMAINED **SIGNIFICANTLY LOWER** THAN THAT OF SEVERAL SMALLER COUNTRIES.

CONCLUSION — KEY TAKEAWAYS

SUMMARY OF FINDINGS

- **COVID IMPACT VARIED SIGNIFICANTLY ACROSS COUNTRIES, ESPECIALLY WHEN ANALYZED USING PERCENTAGE-BASED METRICS RATHER THAN ABSOLUTE COUNTS.**
- **COUNTRY-LEVEL EXTREMES IN DEATH AND INFECTION RATES BECAME MORE STABLE WHEN VIEWED AT THE CONTINENT LEVEL.**
- **TIME-BASED ANALYSIS REVEALED THAT PANDEMIC SEVERITY PEAKED DURING MID-2020 AND DECLINED STEADILY THEREAFTER.**
- **INDIA FOLLOWED A SIMILAR GLOBAL TREND, WITH BOTH DEATH PERCENTAGE AND POPULATION INFECTION RATE REDUCING OVER TIME.**
- **POPULATION-NORMALIZED ANALYSIS HELPED PROVIDE FAIR COMPARISONS ACROSS COUNTRIES WITH VERY DIFFERENT POPULATION SIZES.**